

L7

Bio-inspired Crossmodal Processing

• Intro

- Modalities: Vision, Audio, Haptics, etc
- Multisensory Integration (MSI) yields a coherent, robust and efficient interaction with the environment
- Embedding of MSI in artificial agents and robots acting in the world

- disambiguation of object properties
- Increase accuracy and deal with redundancy in noisy sensory events

• Terminology: depend on discipline and perspective

- Crossmodal Integration: Interdisciplinary
 - Behavioral tasks involving one or several senses
- Multimodal Integration: AI, Intelligent Systems
 - Integration of multiple knowledge-based modalities
- Multisensory Integration: Cognitive Neuroscience
 - Activity of neurons responding to one sense or more
 - Neural processes involved in synthesizing information from different stimuli
- Multisensory Fusion: Engineering, Robotics
 - Mapping several objects to a single object
 - Integration: Combining information into larger system

• Multisensory Integration (MSI)

the process by which information from different sensory modalities is combined to yield a rich, coherent representation of an object or event in the environment.

→ More powerful than just using the most appropriate modality

• Ventriloquism Effect

→ Visual influence on auditory perception

→ Ventriloquism Effect: Visual capture of talking puppet causes perception of sound direction to change

• Modality appropriateness hypothesis

Suppose

→ Spatial Ventriloquism: Localization at position of visual event

ϕ_{aud}

→ Temporal Ventriloquism: flash closer in time than in reality

ϕ_{vis}

→ Double-Flash Illusion: 2 flashes and 2 noise bursts perceived although only one flash

Modality appropriateness

→ Hypothesis: That modality is given full credence which is not appropriate for the stimulus. Attention

• Alternative Hypothesis: Integration

→ Small errors in visual localization and large errors in auditory localization lead to optimal localization being very close to the visual estimate, hence the ventriloquism effect

• Integration Types in NN

Simple Types

Concatenation



Multiplication



Sum



Function

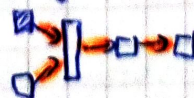


Integration Paths

Late Integration



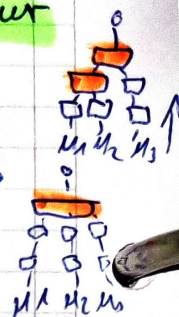
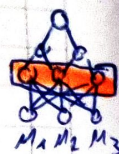
Early Integration



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• NN fusion strategies

- Early fusion strategies take as input a concatenated vector
- Intermediate fusion strategies, first representations of the data are learned, afterwards the modalities are fused. This can occur in one layer or gradually.
- Late fusion strategies combine decisions by sub-models for each modality



• Multisensory Integration: Subcortical Midbrain

- Superior Colliculus: Multisensory, localizes stimuli, present in all vertebrates, motor output

• Cortical Multisensory Integration: Superior Temporal Sulcus

Auditory Cortex

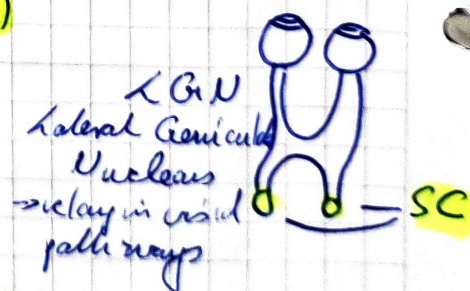
Visual Cortex

Superior Temporal Sulcus

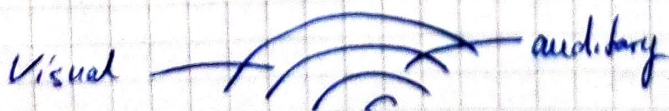


• Bio-inspired MSI in Midbrain (Superior Colliculus)

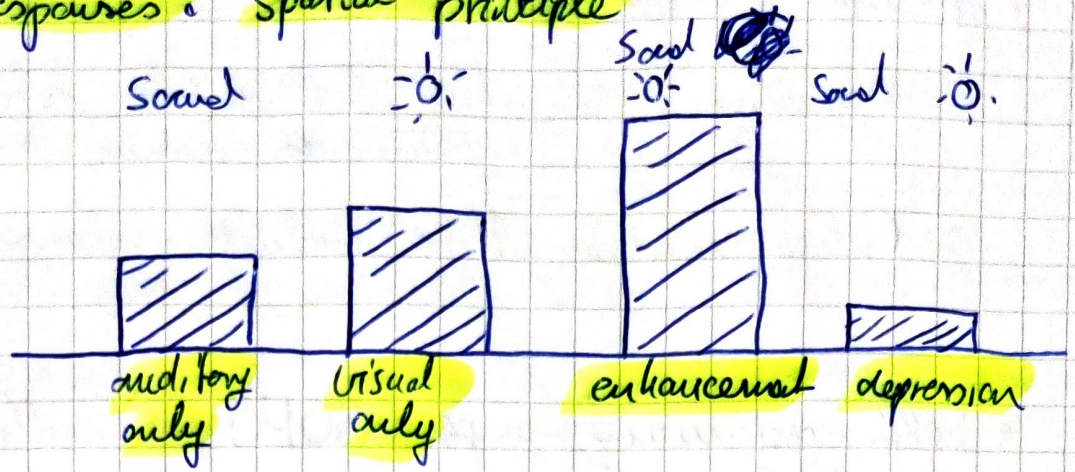
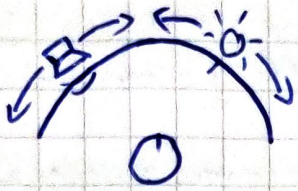
- Primary input is visual, in addition to auditory, somatosensory input
- Visual input to LGN to SC
- SC main visual brain region
- SC sit right over IC (Inferior Colliculus (Sound localization))
- SC is topographically organized: depending on where a stimulus is, it generates activity in a specific column of the SC
- Different sensory modalities to different layers of the SC



Visual down up SC processed



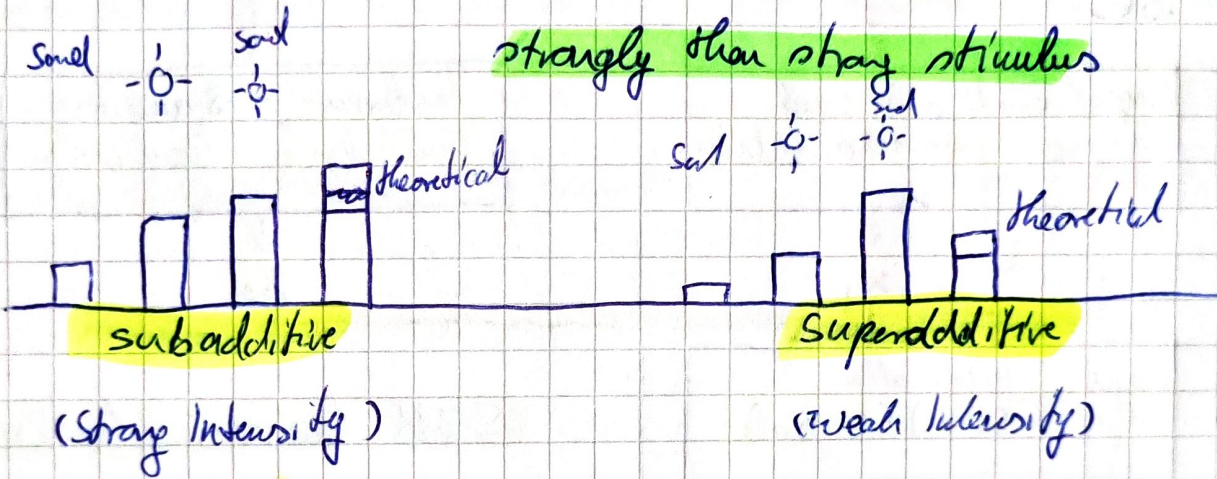
• SC neural responses: spatial principle



→ Enhancement: Visual and auditory stimuli close in time and space

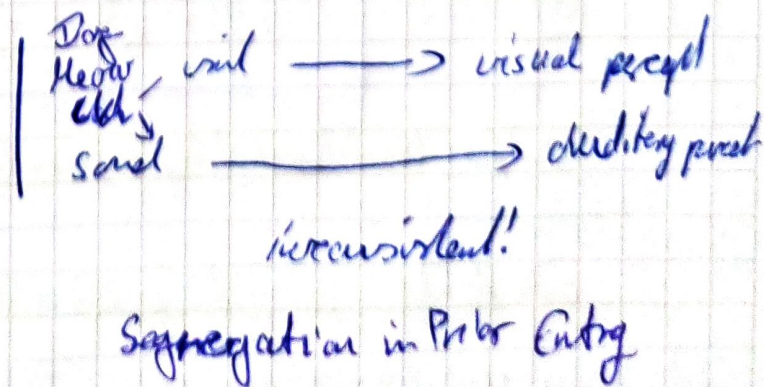
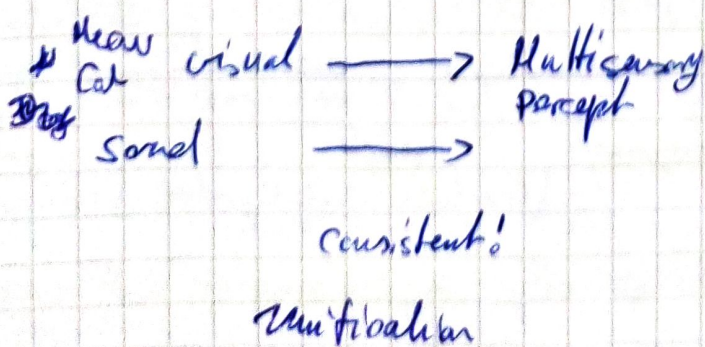
→ Depression: Visual and auditory stimuli displaced from one another

→ Inverse Effectiveness: weak stimuli enhance each other more strongly than strong stimuli



• Unity Assumption

Whenever two or more sensory inputs are perceived as being highly consistent, observers will be more likely to treat them as referring to a single multisensory percept with a common spatiotemporal origin



Neural Phenomena

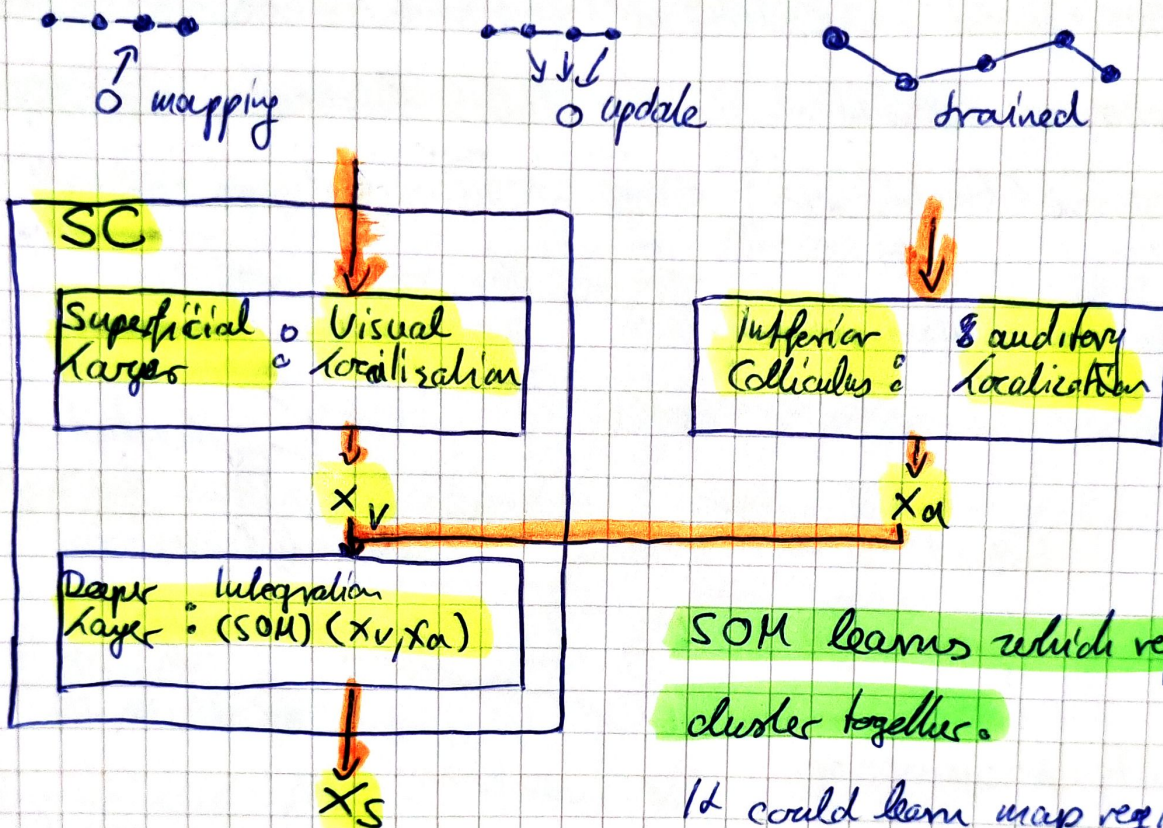
→ Depression $S \downarrow$

More Probable $\frac{1}{2}$ response to only visual than $\frac{1}{2}$ response to both (cross-modal)

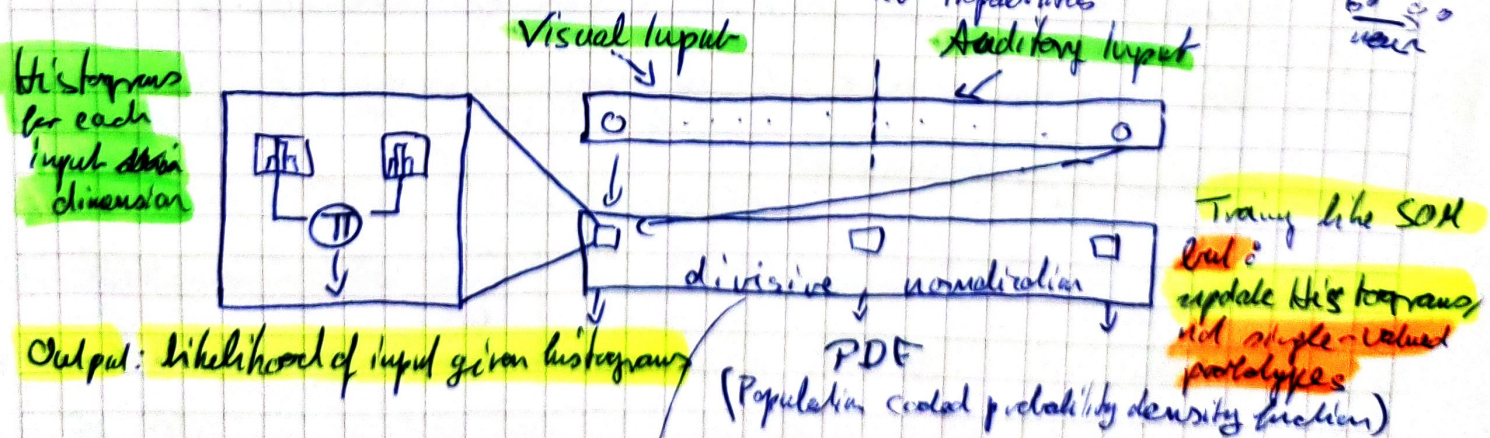
→ Enhancement $S \uparrow$

More Probable response to cross-modal than response to only visual

Self-organizing-maps (SOM) for multimodal integration



Implementation as histogram-based SOM



ANN model combines self-organization and statistical integration

→ UN integrators compatible with ML estimator model

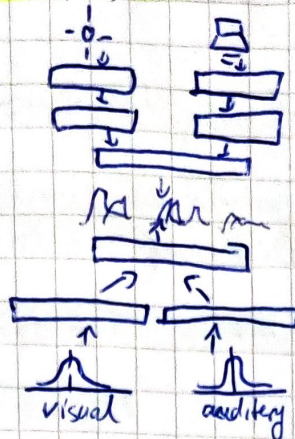
→ The model reproduces some phenomena in biological SC

Depression, spatial principle; Enhancement, increase effectiveness; ~~MS~~

- From SC to Cerebrum - collicular Arch

Top down: Cortical

Bottom-up: collicular subcortical alignment



- Multimodal highest activation for congruent audiovisual patterns

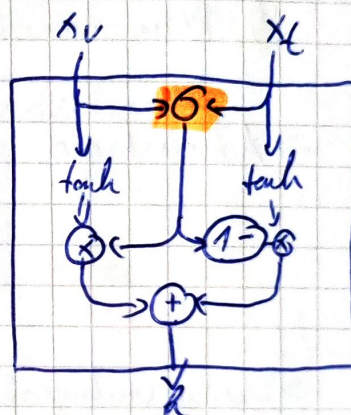
- top-down modulatory projection to bias the integration at subcortical layer

- Integration at unit level: Gated Multimodal Units

→ Information fusion: creates a new representation out of different modalities

→ Information flow is controlled by gate neurons G , that determine modality contribution

→ Fully differentiable NN unit



Summary

→ Models for bio-inspired sound source localization, visual localization and multisensory integration

→ Effectiveness of bio-inspired models

→ Demonstrated behavioral, neurophysiological similarity between models and biological systems

- Levels of neural activity in the minimal layer may provide the reliability of each modality
- Multimodal neurons exhibiting the highest activation for congruent audiovisual patterns
- Activity driven levels of congruency used as top-down modulatory projections to bias the integration at the subcortical layer
- Increase Effectiveness

Superior Colliculus

→ HSI in Brain